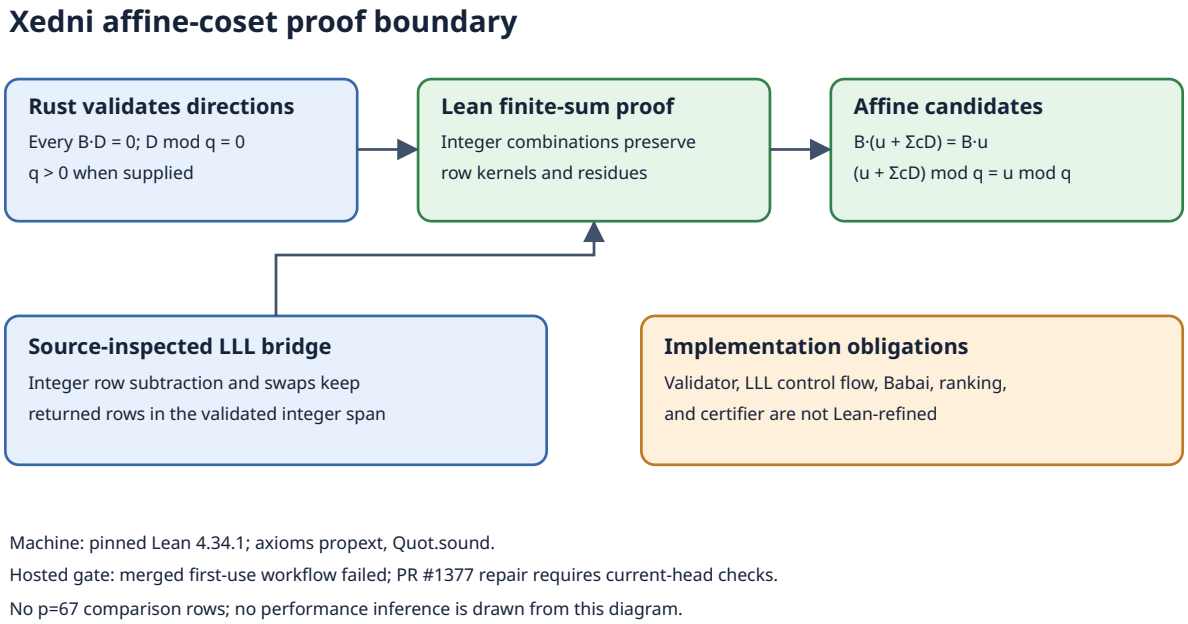
Xedni affine-coset Lean review

Technical packet, 2 November 2026

Scoped result. For finite integer row and direction families, if each direction annihilates every incidence row and is coordinatewise zero modulo positive q, every integer recombination preserves those premises. Each affine candidate $u + \sum_i c_i D_i$ has the same row dot products and residues as u. Three named Lean declarations pass the pinned Lean 4.34.1 checker; their axiom lists contain only `propext` and `Quot.sound`.



Independent review. Blind algebra task `TASK-20261102-1a2b3c` re-derived the finite-sum invariant without producer source, including separate failed controls for the row-kernel and residue premises. Source task `TASK-20261102-4d5e6f` traced the Rust validation, candidate loop, integer LLL row operations, certifier, receipt, and workflow. Both filed separate attestations; the shared independence checker found zero declared violations after a documented cross-repository field-name adapter.

Workflow finding. The merged proof workflow at cryptanalysis `749d807` failed on first-use elan setup even though Lean returned zero. Repair PR #1377 adds explicit toolchain download and binds `lattice.rs` in the source hash and workflow paths. Hosted run 41505647909 passed at earlier repair head `196aa0d`; current-head and post-merge checks are required before describing the merged gate as green.

Claim separation. The Lean algebra is machine checked and faithful to the conditional candidate invariant. The Rust LLL span property is source-inspected, not a formal refinement. Parser behavior, full validator and LLL control flow, Babai choice, ranking, and certifier remain distinct implementation obligations. The frozen $p = 67$ diagnostic has no run rows because checkpoint upload was denied, so no performance curve is inferred.

Source bindings. Producer commit `d243c3c`; Lean SHA-256 `c39148ea...e757`; Rust candidate SHA-256 `b08c9484...ac96`; LLL SHA-256 `0687b6c0...0b`. The full hashes, per-claim table, controls, and report links are in the adjacent `README.md`. The cross-field degree graph was checked against its CSV and left unchanged because this proof changes no graph input.